

Shyamendra Singh

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EDUCATION

B.Tech (Agricultural Engineering)	Dayalbagh Educational Institute, Agra	8.02 CGPA (Till 5 th Sem)	2026 (Pursuing)
Intermediate (XII)	SMT Maharaj Kunwari Inter College, Agra	80.2%	2021
Highschool (X)	SMT Maharaj Kunwari Inter College, Agra	86.5%	2019

PROFESSIONAL EXPERIENCE

Virtual Labs, MoE-NMEICT	July 2023 – Present
• <i>Project Intern</i>	<i>Agra, UP</i>
○ Co-operating as an intern in Virtual Labs, a Ministry of Education project under NMEICT, at my college.	
○ Responsible for designing simulations, illustrations, and preparing content of the virtual experiments.	
ICAR - Indian Agricultural Research Institute (IARI)	June 2024
• <i>Internship Trainee</i>	<i>Pusa, New Delhi</i>
○ Gained expertise in mechatronics, CAD and 3D printing, applying these skills to prototype tools and equipments for agricultural applications. Developed <i>Wireless BMI Measuring Device</i> .	
Hankernest Technologies Pvt. Ltd.	June 2022
• <i>Internship Trainee</i>	<i>Dharamkot, HP</i>
○ A training-cum-internship program covering python programming, practical AI/ML (transfer learning system), practical electronics, internet of things, 3D printing, and CAD designing.	

TECHNICAL SKILLS & INTERESTS

- **CAD:** AutoCAD, SolidWorks
- **Programming & Scripting:** Python, HTML/CSS, GitHub
- **Spatial Analysis:** QGIS, ArcGIS
- **Other Tools:** Arduino IDE, Microsoft Office, Canva, Figma
- **Interests:** 3D Designing, GIS for Agriculture, Additive Manufacturing, IoT, Machine Learning

PROJECTS

VARDAN – An AI Powered Smart Spraying Equipment (AI, Computer Vision, ESP32, App Development)
A low-cost smart spraying equipment utilising AI, computer vision and IoT for automatic weed detection and chemical spraying.
Wireless BMI Measuring Device (3D CAD & Printing, ESP32, Arduino IDE)
A portable device for wireless measurement of anthropometric dimensions and body mass index using sensors and microcontrollers, advancing ergonomic studies in agricultural settings.
Crop Recommendation Web App  (Python, KNN, Flask)
A web application utilizing the k-nearest neighbors (KNN) algorithm for crop recommendation based on soil and climatic parameters, aimed at enhancing decision-making for farmers.
Soil Texture Calculator  (Python, Matplotlib, Flask)
A web-based tool for soil classification according to USDA standards, providing automated textural analysis and generating ternary graphs for visual representation.

CERTIFICATIONS

- **Supervised Machine Learning: Regression and Classification** course from DeepLearning.AI
- **Artificial Intelligence for Everyone (AI4E)** course from Dayalbagh Educational Institute, Agra
- **Rajasthan State Certificate in Information Technology (RSCIT)** from VMOU, Kota

HACKATHONS & WORKSHOPS

- Qualified the institution level stage of **Smart India Hackathon (SIH 2023, SIH2024)**
- Participated in one day online workshop on **Remote Sensing based Data Analytics in Agriculture** organised by IIRS
- Participated in one-week hands-on workshop on **Digital Manufacturing** organised by Dayalbagh Educational Institute, Agra